1.1.3 ▶ ▶ The Projectile Trajectory Equation

All trajectories of objects in projectile motion are parabolas. Sometimes, it is useful to know the equation of these trajectories, as they can give us valuable insight into the maximum range of a projectile or the envelope of safety for a projectile (we will go over the former of these examples since the uses of this equation are not as immediately apparent as other equations derived here)[2].

We start by considering a projectile launched with a speed of v_0 at an angle of θ above the horizontal at the origin and at $t = 0$. At a certain time t, the x coordinate of this projectile is

$$x = v_0 \cos\theta \cdot t \tag{1.14}$$

since $x = v_x t$, and the y coordinate is

$$y = v_0 \sin\theta \cdot t - \tfrac{1}{2} g t^2 \tag{1.15}$$

using eq. 1.4. Substituting t from eq. 1.14 into eq. 1.15, we have

$$\boxed{y = \tan\theta \cdot x - \frac{g \sec^2\theta}{2v_0^2} \cdot x^2}. \tag{1.16}$$

The trick to use this equation is rewriting the $\sec^2\theta$ as $1 + \tan^2\theta$ using a Pythagorean identity. Making this substitution yields the following quadratic in $\tan\theta$:

$$-\frac{gx^2}{2v_0^2}(1 + \tan^2\theta) + x\tan\theta - y = 0$$

$$1 + \tan^2\theta - \frac{2v_0^2}{gx}\tan\theta + \frac{2v_0^2 y}{gx^2}$$

$$\tan^2\theta - \frac{2v_0^2}{gx}\tan\theta + \frac{2v_0^2 y}{gx^2} + 1 = 0 \tag{1.17}$$

Where the expression has been divided through to remove any coefficient in front of the $\tan^2\theta$ term.

Since this is a quadratic equation, it can be maximized by setting the discriminant of the quadratic formula to zero. Using this optimization, we can find the maximum range R_{max} of a projectile launched at *any* height H.

Let's say that a projectile is launched from a height H at an optimal angle θ_{max} such that it travels a horizontal distance R_{max}. If we plug all of these values into eq. 1.17, we get [3].

$$\tan^2\theta - \frac{2v_0^2}{gR_{max}}\tan\theta - \frac{2v_0^2 H}{gR_{max}^2} + 1 = 0. \tag{1.18}$$

[2] Although not obvious, the use of this technique has been featured multiple times on physics competitions and thus is important :)

[3] Note that the projectile travels a distance $-H$ in the vertical direction, so $y = -H$

Since the values are all optimized, the discriminant ($\sqrt{b^2 - 4ac}$) must equal zero. Writing this condition, we have

$$\frac{4v_0^4}{g^2 R_{max}^2} - 4\left(1 - \frac{2v_0^2 H}{g R_{max}^2}\right)$$

$$\frac{v_0^4}{g^2} = R_{max}^2 - \frac{2v_0^2 H}{g}$$

$$R_{max}^2 = \frac{v_0^2}{g}\left(\frac{v_0^2}{g} + 2H\right). \tag{1.19}$$

We can find θ_{max} using the quadratic formula, and since the discriminant is zero, the quadratic formula reduces down to $\tan\theta_{max} = -b/2a$. So,

$$\tan\theta_{max} = \frac{v_0^2}{gR_{max}}. \tag{1.20}$$

If we set $H = 0$ in eq. 1.18 and 1.19 (just a projectile launched on a flat surface), we get $R_{max} = v_0^2/g$ and $\theta_{max} = 45°$ as predicted in eq. 1.13 ★